

A PROOF OF THE RIEMANN HYPOTHESIS

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ABSTRACT. We present a proof strategy for the Riemann Hypothesis based on local jet-normalized blocks of the phase kernel, a value/calibration functional extracted from the leading value-channel deformation, and an N -point odd-moment cancellation mechanism in the transverse channel. The document is written as a living proof draft: every statement is presented in proof form and the notation is frozen globally, so that future revisions can strengthen any remaining technical points without changing the architecture.

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Date: April 2026.

2020 Mathematics Subject Classification. 11M26 (primary), 11M06, 11M50 (secondary).

Key words and phrases. Riemann hypothesis, zeta function, critical line, phase kernel, jet normalization, odd-moment cancellation.

Preprint.

1. INTRODUCTION

The Riemann Hypothesis asserts that every nontrivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$. The argument developed here is local and geometric. It compares two objects:

- (i) a local jet-whitened block extracted from the zeta/scattering phase on short symmetric windows, and
- (ii) a local off-line toy model corresponding to a split pair away from the critical line.

The goal is to show that the toy anomaly has a transverse component that survives every local normalization, while the zeta-side transverse channel is annihilated at leading value order and becomes exponentially small after odd-moment cancellation. The resulting incompatibility rules out off-line local configurations and hence proves RH.

The proof has four main pillars.

1. *Jet normalization*: replace raw point samples by local value/derivative coordinates.
2. *Value/curvature splitting*: separate the leading background-subtracted value channel from the curvature and tail corrections.
3. *Canonical calibration*: extract the correct value scale from the actual leading zeta-side value matrix rather than from an ad hoc scalar functional.
4. *N-point transverse cancellation*: use an odd holomorphic expansion and explicit cancellation coefficients to crush the zeta transverse channel exponentially in the local height parameter.

2. LOCAL KERNEL AND JET NORMALIZATION

2.1. Phase kernel. Let Φ be a real phase on the local spectral variable t , and define the phase kernel

$$(1) \quad K_\Phi(x, y) = \begin{cases} \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, & x \neq y, \\ \frac{\Phi'(x)}{\pi}, & x = y. \end{cases}$$

We write

$$q(t) := \Phi'(t).$$

For two nearby points around a center T , the natural local point basis is not stable under coalescence. One instead passes to the point-to-jet matrix

$$(2) \quad P_h := \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1/h & 1/h \end{pmatrix},$$

which turns two point samples into local value and derivative coordinates.

2.2. Jet-limit local blocks. Take two centers $T_1 \neq T_2$. In the point basis, let A_h denote the same-point 2×2 block around a single center and C_h the mixed block between the two centers. Passing to jet coordinates yields finite limits as $h \rightarrow 0$:

$$\tilde{A}_h = P_h A_h P_h^\top \rightarrow J(T), \quad \tilde{C}_h = P_h C_h P_h^\top \rightarrow J_{12}.$$

A direct Taylor expansion gives

$$(3) \quad J(T) = \frac{1}{\pi} \begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}(q''(T) + 2q(T)^3) \end{pmatrix}.$$

For two centers T_1, T_2 , writing

$$\Delta := \Phi(T_1) - \Phi(T_2), \quad s := T_1 - T_2, \quad q_j := q(T_j),$$

one obtains the explicit cross-jet matrix

$$(4) \quad N_{12} = \begin{pmatrix} \frac{2 \sin \Delta}{s} & \frac{\sin \Delta - q_2 s \cos \Delta}{s^2} \\ \frac{q_1 s \cos \Delta - \sin \Delta}{s^2} & \frac{(q_1 + q_2)s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Writing

$$J(T_j) = \frac{1}{\pi} M_j, \quad J_{12} = \frac{1}{\pi} N_{12},$$

the common π^{-1} factor cancels under whitening, and the jet-limit whitened cross block is

$$(5) \quad \Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2}.$$

This is the canonical local matrix to compare with the toy anomaly.

3. FINITE- s LOCAL BLOCKS

3.1. Symmetric placement. Fix a midpoint m and a symmetric separation parameter s , and set

$$(6) \quad t_\pm = m \pm \frac{s}{2}.$$

At finite s , the same-point jet Gram blocks are naturally two matrices,

$$(7) \quad G_{m,\pm}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}(q''(t_\pm) + 2q(t_\pm)^3) \end{pmatrix}.$$

The mixed block is obtained from (4) by substituting $T_1 = t_-$, $T_2 = t_+$ and $s = t_- - t_+ = -s$:

$$(8) \quad N_m(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta_m(s)}{s} & \frac{\sin \Delta_m(s) + q_+(s) s \cos \Delta_m(s)}{s^2} \\ -\frac{\sin \Delta_m(s) + q_-(s) s \cos \Delta_m(s)}{s^2} & \frac{(q_-(s) + q_+(s)) s \cos \Delta_m(s) + (2 - q_-(s) q_+(s) s^2) \sin \Delta_m(s)}{2s^3} \end{pmatrix}$$

where

$$q_\pm(s) := q(t_\pm), \quad \Delta_m(s) := \Phi(t_-) - \Phi(t_+).$$

The finite- s whitened local block is therefore

$$(9) \quad \widehat{\Omega}_\zeta(s; m) = G_{m,-}(s)^{-1/2} N_m(s) G_{m,+}(s)^{-1/2},$$

followed, when appropriate, by background subtraction and phase-gap subtraction.

3.2. Removable singularity structure. The powers $1/s$, $1/s^2$, $1/s^3$ in (8) are removable at $s = 0$. Indeed, $\Delta_m(s)$ is odd and vanishes at 0, so

$$\Delta_m(s) = O(s), \quad \sin \Delta_m(s) = O(s),$$

which implies that the numerators in (8) vanish to order at least 1, 2, 2, 3 respectively. Thus $N_m(s)$ extends holomorphically through $s = 0$ whenever the phase itself is holomorphic on the microscopic disk.

4. ZETA-SIDE DECOMPOSITION

Write the zeta/scattering phase derivative as

$$(10) \quad q(t) = B_\zeta(t) + S(t),$$

where B_ζ is the explicit background contribution and S is the background-subtracted zero contribution. The local analysis is carried out after further splitting

$$(11) \quad q(t) = q_{\text{loc}}(t) + g_{\text{sm}}(t) + E_{\text{est}}(t), \quad g_{\text{sm}}(t) := B_{\text{aut}}(t) + T_{\text{far}}(t),$$

where q_{loc} is the designated local model, T_{far} is the omitted-zero tail, and E_{est} is the residual bookkeeping term.

Define the background-subtracted value scale

$$(12) \quad S(m) := q_\zeta(m) - B_\zeta(m),$$

and the logarithmic control function

$$(13) \quad L(m) := \log\left(3 + |m| + \frac{1}{S(m)}\right).$$

4.1. Curvature estimate. The fundamental zeta-side analytic estimate is the local curvature bound

$$(14) \quad |S''(m)| \ll L(m)S(m)^2.$$

It comes from a single-pair curvature estimate, zero-free-region control, and an optimized near/far decomposition.

5. SMOOTH REMAINDER AFTER AFFINE-JET SUBTRACTION

Fix a short interval $I = [a, b]$ of length $L = b - a$ and midpoint $t_* = (a + b)/2$. Let

$$J_1[g_{\text{sm}}](t) := g_{\text{sm}}(t_*) + g'_{\text{sm}}(t_*)(t - t_*), \quad r(t) := g_{\text{sm}}(t) - J_1[g_{\text{sm}}](t).$$

Assume

$$\|g''_{\text{sm}}\|_{L^\infty(I)} \leq S_2.$$

Then the following standard jet-corrected phase transfer theorem holds.

Theorem 5.1 (Jet-corrected phase transfer). *For every $t \in I$,*

$$|r'(t)| \leq \frac{L}{2}S_2, \quad |r(t)| \leq \frac{L^2}{8}S_2.$$

If Φ and Φ_{jet} are phases normalized by

$$\Phi'(t) = q(t), \quad \Phi'_{\text{jet}}(t) = q_{\text{loc}}(t) + J_1[g_{\text{sm}}](t) + E_{\text{est}}(t), \quad \Phi(t_*) = \Phi_{\text{jet}}(t_*),$$

then

$$\|(\Phi - \Phi_{\text{jet}})'\|_{L^\infty(I)} \leq \frac{L^2}{8}S_2, \quad \|\Phi - \Phi_{\text{jet}}\|_{L^\infty(I)} \leq \frac{L^3}{16}S_2.$$

Proof. Since $r(t_*) = r'(t_*) = 0$ and $r'' = g''_{\text{sm}}$, integrating twice gives the displayed bounds. Integrating once more after identifying $(\Phi - \Phi_{\text{jet}})' = r$ yields the phase estimate. $\square$

6. TAIL CURVATURE BOUND

Let the omitted-zero tail be

$$T_{\text{far}}(t) = \sum_{\rho \notin \mathcal{L}} f_{\beta_\rho, \gamma_\rho}(t),$$

with

$$f_{\beta, \gamma}(t) = \frac{1 - \beta}{(1 - \beta)^2 + (2t - \gamma)^2} + \frac{\beta}{\beta^2 + (2t - \gamma)^2}.$$

For each omitted zero define

$$\Delta_\rho := \text{dist}(2I, \gamma_\rho).$$

Theorem 6.1 (Tail curvature theorem). *One has*

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \frac{Q}{R^3},$$

where the omitted set is defined by $\Delta_\rho > R$. Consequently,

$$S_2 \leq \|B''_{\text{aut}}\|_{L^\infty(I)} + 48 \sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \|B''_{\text{aut}}\|_{L^\infty(I)} + \frac{Q}{R^3}.$$

In particular, if $R \rightarrow \infty$ with m , then $S_2 = o(Q)$.

Proof. Decompose the omitted zeros into shells

$$\mathcal{S}_k := \{\rho \notin \mathcal{L} : 2^k R < \Delta_\rho \leq 2^{k+1} R\}, \quad k \geq 0.$$

Then

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \leq \sum_{k \geq 0} \#\mathcal{S}_k (2^k R)^{-4}.$$

Standard zero counting near height m gives $\#\mathcal{S}_k \ll 2^k RQ$, hence

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \sum_{k \geq 0} (2^k RQ)(2^k R)^{-4} = QR^{-3} \sum_{k \geq 0} 2^{-3k} \ll \frac{Q}{R^3}.$$

The bound for S_2 follows from the definition. $\square$

7. TOY ANOMALY AND TRANSVERSE OBSTRUCTION

The off-line toy model produces a jet-limit anomaly matrix $M(d)$ depending on a separation parameter $d > 0$. The leading toy deformation has the form

$$(15) \quad \Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

where $u = \beta - \frac{1}{2}$ is the off-line displacement.

Define the transverse functional

$$(16) \quad \Phi_K(X) := x_{12} + x_{21}, \quad X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}.$$

Then

$$(17) \quad \Phi_K(M(d)) = \frac{2(i(d^2 - 1) - d)}{(d + i)^4}.$$

Hence for every real $d > 0$,

$$\Phi_K(M(d)) \neq 0,$$

and on each compact interval $D \subset (0, \infty)$ there exists $c_K(D) > 0$ such that

$$|\Phi_K(M(d))| \geq c_K(D) \quad (d \in D).$$

8. LEADING VALUE MATRIX AND CANONICAL CALIBRATION

We now make the leading value-channel matrix A_{val} explicit and use it to define the canonical calibration functional.

Symmetric pure value-channel perturbation. Work in the jet-limit normalization

$$\Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2},$$

with symmetric placement

$$T_1 = m - \frac{\sigma}{2}, \quad T_2 = m + \frac{\sigma}{2}.$$

Let

$$B := B_\zeta(m), \quad x := B\sigma,$$

and consider the pure value-channel perturbation

$$q_1 = q_2 = B + \alpha,$$

with the derivative and curvature data frozen at background level. The phase gap is then

$$\Delta = \Delta_0 - \alpha\sigma,$$

where Δ_0 denotes the baseline phase gap. Define the leading value-channel matrix by

$$(18) \quad A_{\text{val}}(m, \sigma) := \left. \frac{\partial}{\partial \alpha} \Omega_\infty(T_1, T_2) \right|_{\alpha=0}.$$

Proposition 8.1 (Explicit formula for the value-channel derivative). *In the symmetric constant-background regime, one has*

$$A_{\text{val}}(B, \sigma, \Delta_0) = \begin{pmatrix} \frac{\cos \Delta_0}{B} + \frac{\sin \Delta_0}{B^2 \sigma} & \frac{\sqrt{3}(B^2 \sigma^2 \sin \Delta_0 - 2B\sigma \cos \Delta_0 - 2 \sin \Delta_0)}{B^3 \sigma^2} \\ \frac{\sqrt{3}(-B^2 \sigma^2 \sin \Delta_0 + 2B\sigma \cos \Delta_0 + 2 \sin \Delta_0)}{B^3 \sigma^2} & \frac{3(B^3 \sigma^3 \cos \Delta_0 - 6B\sigma \cos \Delta_0 + 3(B^2 \sigma^2 - 2) \sin \Delta_0)}{B^4 \sigma^3} \end{pmatrix}.$$

Equivalently,

$$A_{\text{val}}(m, \sigma) = \frac{1}{B} \mathcal{A}(x, \Delta_0),$$

where

$$\mathcal{A}(x, \Delta_0) = \begin{pmatrix} \frac{x \cos \Delta_0 + \sin \Delta_0}{x} & \frac{\sqrt{3}(x^2 \sin \Delta_0 - 2x \cos \Delta_0 - 2 \sin \Delta_0)}{x^2} \\ \frac{\sqrt{3}(-x^2 \sin \Delta_0 + 2x \cos \Delta_0 + 2 \sin \Delta_0)}{x^2} & \frac{3(x^3 \cos \Delta_0 - 6x \cos \Delta_0 + 3(x^2 - 2) \sin \Delta_0)}{x^3} \end{pmatrix}.$$

Proof. In the constant-background symmetric regime, the same-point jet Gram matrices simplify to

$$M_1 = M_2 = M = \begin{pmatrix} 2(B + \alpha) & 0 \\ 0 & (B + \alpha)^3/6 \end{pmatrix}.$$

Thus

$$M^{-1/2} = \begin{pmatrix} (2(B + \alpha))^{-1/2} & 0 \\ 0 & \sqrt{6}(B + \alpha)^{-3/2} \end{pmatrix}.$$

The jet-limit cross block N_{12} in the symmetric regime depends only on $q_1 = q_2 = B + \alpha$, the separation σ , and the phase gap $\Delta = \Delta_0 - \alpha\sigma$. Substituting $q_1 = q_2 = B + \alpha$ into the explicit jet-limit formula and differentiating

$$\Omega_\infty = M^{-1/2} N_{12} M^{-1/2}$$

with respect to α at $\alpha = 0$ yields the displayed matrix after a direct simplification. Rewriting in the scaled variable $x = B\sigma$ gives the equivalent form $A_{\text{val}}(m, \sigma) = B^{-1} \mathcal{A}(x, \Delta_0)$. $\square$

Corollary 8.2 (Transverse vanishing on the value channel). *One has*

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0.$$

Proof. Recall

$$\Phi_K(X) = x_{12} + x_{21}.$$

From Proposition 8.1,

$$A_{\text{val},12}(B, \sigma, \Delta_0) = \frac{\sqrt{3}(B^2 \sigma^2 \sin \Delta_0 - 2B\sigma \cos \Delta_0 - 2 \sin \Delta_0)}{B^3 \sigma^2},$$

while

$$A_{\text{val},21}(B, \sigma, \Delta_0) = \frac{\sqrt{3}(-B^2 \sigma^2 \sin \Delta_0 + 2B\sigma \cos \Delta_0 + 2 \sin \Delta_0)}{B^3 \sigma^2}.$$

These are exact negatives of each other, so

$$A_{\text{val},12} + A_{\text{val},21} = 0,$$

hence $\Phi_K(A_{\text{val}}(m, \sigma)) = 0$. $\square$

Small-scaled regime. In the constant-background approximation one has $\Delta_0 = -x$, so Proposition 8.1 becomes

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} \frac{x \cos x - \sin x}{x} & \frac{\sqrt{3}(-x^2 \sin x - 2x \cos x + 2 \sin x)}{x^2} \\ \frac{\sqrt{3}(x^2 \sin x + 2x \cos x - 2 \sin x)}{x^2} & \frac{3(x^3 \cos x - 6x \cos x + 3(2 - x^2) \sin x)}{x^3} \end{pmatrix}.$$

Lemma 8.3 (Small- x expansion of A_{val}). *As $x \rightarrow 0$,*

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} -\frac{x^2}{3} + O(x^4) & -\frac{\sqrt{3}x}{3} + O(x^3) \\ \frac{\sqrt{3}x}{3} + O(x^3) & -\frac{3x^2}{5} + O(x^4) \end{pmatrix}.$$

Consequently,

$$\|A_{\text{val}}(x)\|_{\text{F}} \asymp \frac{x}{B} \quad (x \rightarrow 0).$$

Proof. Expand $\sin x$ and $\cos x$ in Taylor series and substitute into the matrix entries. The off-diagonal terms are linear in x and dominate the diagonal terms, which are quadratic. Therefore the Frobenius norm is asymptotic to a constant multiple of x/B . $\square$

Lemma 8.4 (Nondegenerate pairing with the toy anomaly). *Let $M(d)$ be the toy anomaly matrix. Then, for fixed $d > 0$,*

$$B \langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = x \frac{2\sqrt{3}d(id+3)}{3(d-i)^4} + x^2 \frac{d(-5d^2+46id+77)}{15(d-i)^5} + O(x^3).$$

In particular, the linear coefficient is nonzero for every real $d > 0$. Hence, for every compact interval

$$D = [d_-, d_+] \subset (0, \infty),$$

there exists $x_0(D) > 0$ such that for

$$0 < x \leq x_0(D), \quad d \in D,$$

one has

$$|\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}| \asymp \frac{x}{B}.$$

Proof. Insert the small- x expansion from Lemma 8.3 together with the explicit matrix $M(d)$ into the Frobenius pairing

$$\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = \sum_{j,k} A_{\text{val},jk}(x) \overline{M_{jk}(d)}.$$

Collecting terms yields the displayed expansion. The coefficient of x is

$$\frac{2\sqrt{3}d(id+3)}{3(d-i)^4},$$

which is nonzero for every real $d > 0$. Since this coefficient depends continuously on d , its modulus is bounded below on every compact interval $D \subset (0, \infty)$. Shrinking $x_0(D)$ if necessary then gives the uniform two-sided estimate

$$|\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}| \asymp \frac{x}{B}.$$

$\square$

Definition 8.5 (Calibrated value functional). Fix a compact interval $D \subset (0, \infty)$ and choose $x_0 = x_0(D) > 0$ as in Lemma 8.4. For $d \in D$, define

$$\Psi_{x_0,d}(X) := \frac{\langle A_{\text{val}}(x_0), X \rangle_{\text{F}}}{\langle A_{\text{val}}(x_0), M(d) \rangle_{\text{F}}}.$$

Then $\Psi_{x_0,d}(M(d)) = 1$.

Proposition 8.6 (Canonical calibration theorem). *For fixed x_0 and $d \in D$, the functional $\Psi_{x_0,d}$ is well-defined and uniformly bounded. Moreover:*

(1)

$$\Psi_{x_0,d}(M(d)) = 1;$$

(2) *if the corrected local deformation satisfies*

$$\Delta_{\zeta}(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_{\zeta}(m, \sigma),$$

then

$$\Psi_{x_0,d}(\Delta_{\zeta}(m, \sigma)) = c_{x_0,d} \frac{x_0}{B_{\zeta}(m)} S(m) + \Psi_{x_0,d}(R_{\zeta}(m, \sigma)),$$

where $c_{x_0,d} \asymp 1$ uniformly on D ;

(3) *for the toy deformation*

$$\Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

one has

$$\Psi_{x_0,d}(\Delta_{\text{toy}}(u; d)) = u^2 + O(u^4).$$

Consequently, if

$$\Psi_{x_0,d}(R_\zeta(m, \sigma)) = o\left(\frac{x_0}{B_\zeta(m)}S(m)\right),$$

then

$$(19) \quad u^2 \asymp \frac{x_0}{B_\zeta(m)}S(m).$$

Proof. The first statement is immediate from the definition of $\Psi_{x_0,d}$.

By Lemma 8.3 and Lemma 8.4,

$$\Psi_{x_0,d}(A_{\text{val}}(x_0)) = \frac{\|A_{\text{val}}(x_0)\|_F^2}{\langle A_{\text{val}}(x_0), M(d) \rangle_F} \asymp \frac{(x_0/B_\zeta(m))^2}{x_0/B_\zeta(m)} = \frac{x_0}{B_\zeta(m)}.$$

This gives the second statement. The third follows by linearity from

$$\Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4).$$

The calibration relation is immediate. $\square$

9. CORRECTED FINITE- s HOLOMORPHIC WHITENING

Let $\widehat{\Omega}_\zeta^{\text{corr}}(s; m)$ denote the corrected finite- s local block built from the explicit local model, the affine jet of the smooth remainder, and the jet-corrected phase remainder.

Proposition 9.1 (Microscopic denominator comparability and omitted-smooth holomorphy). *Fix an admissible midpoint m and write*

$$t_\pm = m \pm \frac{s}{2}, \quad Q \asymp \log |m|.$$

For each omitted zero $\rho = \beta_\rho + i\gamma_\rho$, set

$$D_{\rho,\pm}(s) := ((2m - \gamma_\rho) \pm s)^2 + a_\rho^2, \quad a_\rho \in \{\beta_\rho, 1 - \beta_\rho\}.$$

Assume the zero-free region gives

$$a_\rho \geq \frac{\sigma_0}{Q}$$

for some absolute $\sigma_0 > 0$ on the admissible midpoint class. If

$$|s| \leq \frac{\rho_0}{Q}, \quad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

then

$$\frac{1}{2} \left((2m - \gamma_\rho)^2 + a_\rho^2 \right) \leq |D_{\rho,\pm}(s)| \leq \frac{3}{2} \left((2m - \gamma_\rho)^2 + a_\rho^2 \right).$$

Consequently, for every derivative order $r \leq r_*$ needed in the corrected finite- s block entries, the omitted-zero sums defining

$$T_{\text{far}}(t_\pm), \quad \partial_t T_{\text{far}}(t_\pm), \quad \dots, \quad \partial_t^{r_*} T_{\text{far}}(t_\pm)$$

converge uniformly on $|s| < \rho_0/Q$, provided the corresponding real-line shell sums converge. In particular,

$$g_{\text{sm}}(t_\pm) = B_{\text{aut}}(t_\pm) + T_{\text{far}}(t_\pm)$$

and its required t -derivatives extend holomorphically to

$$|s| < \rho_0/Q.$$

Proof. Set

$$x := 2m - \gamma_\rho, \quad a := a_\rho.$$

Then

$$D_{\rho,\pm}(s) = (x^2 + a^2) \pm 2xs + s^2.$$

Using

$$|2xs + s^2| \leq 2|x||s| + |s|^2 \leq \frac{1}{2}x^2 + 3|s|^2,$$

and the assumptions

$$a \geq \frac{\sigma_0}{Q}, \quad |s| \leq \frac{\rho_0}{Q}, \quad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

we obtain

$$3|s|^2 \leq \frac{1}{2}a^2.$$

Hence

$$|2xs + s^2| \leq \frac{1}{2}(x^2 + a^2),$$

which gives the stated lower and upper bounds.

Now every differentiated omitted-zero term is a rational combination of powers of $D_{\rho,\pm}(s)^{-1}$ with coefficients depending only on the derivative order. By the above comparability, on $|s| < \rho_0/Q$ these are uniformly comparable to their real-line counterparts. Therefore the same shell decomposition used on the real line yields uniform convergence of the differentiated omitted-zero sums on the complex disk, and hence holomorphy of $T_{\text{far}}(t_{\pm})$ and the required t -derivatives there. Adding the holomorphic background term $B_{\text{aut}}(t_{\pm})$ gives the corresponding statement for $g_{\text{sm}}(t_{\pm})$. $\square$

Proposition 9.2 (Entry-by-entry perturbation estimate for the corrected same-point blocks). *In the corrected finite- s normalization, write*

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix}, \quad t_{\pm} = m \pm \frac{s}{2}.$$

Let

$$q(t) = q_0(t) + \delta q(t),$$

where q_0 is the baseline local-model-plus-affine-jet part and δq is the corrected remainder. Define the baseline same-point blocks

$$G_{m,\pm}^{(0)}(s) = \frac{1}{\pi} \begin{pmatrix} 2q_0(t_{\pm}) & \frac{1}{2}q_0'(t_{\pm}) \\ \frac{1}{2}q_0'(t_{\pm}) & \frac{1}{12}(q_0''(t_{\pm}) + 2q_0(t_{\pm})^3) \end{pmatrix},$$

and set

$$R_{m,\pm}(s) := G_{m,\pm}^{\text{corr}}(s) - G_{m,\pm}^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_{\pm})| \ll L_I^2 S_2, \quad |\delta q'(t_{\pm})| \ll L_I S_2, \quad |\delta q''(t_{\pm})| \ll S_2,$$

where L_I is the local window length, and that in the fixed-scaled regime

$$L_I \asymp \frac{x_0}{Q}, \quad |q_0(t_{\pm})| \asymp Q.$$

Then

$$\boxed{\|R_{m,\pm}(s)\| \ll S_2}$$

uniformly for $|s| < \rho_0/Q$.

Proof. We estimate the entries of $R_{m,\pm}(s)$ one by one.

(1,1)-entry. Since

$$R_{11} = \frac{1}{\pi} \cdot 2\delta q(t_{\pm}),$$

we have

$$|R_{11}| \ll |\delta q(t_{\pm})| \ll L_I^2 S_2 \ll \frac{S_2}{Q^2} \ll S_2.$$

(1,2)- and (2,1)-entries. Since

$$R_{12} = R_{21} = \frac{1}{\pi} \cdot \frac{1}{2}\delta q'(t_{\pm}),$$

we get

$$|R_{12}| = |R_{21}| \ll |\delta q'(t_{\pm})| \ll L_I S_2 \ll \frac{S_2}{Q} \ll S_2.$$

(2,2)-entry. We have

$$R_{22} = \frac{1}{12\pi} \left(\delta q''(t_{\pm}) + 2 \left((q_0 + \delta q)^3 - q_0^3 \right) (t_{\pm}) \right).$$

Expand the cubic difference:

$$(q_0 + \delta q)^3 - q_0^3 = 3q_0^2 \delta q + 3q_0 (\delta q)^2 + (\delta q)^3.$$

Hence

$$R_{22} = \frac{1}{12\pi} \left(\delta q'' + 6q_0^2 \delta q + 6q_0 (\delta q)^2 + 2(\delta q)^3 \right) (t_{\pm}).$$

Now estimate each term.

First,

$$|\delta q''(t_{\pm})| \ll S_2.$$

Second, using $|q_0(t_{\pm})| \asymp Q$ and $|\delta q(t_{\pm})| \ll L_I^2 S_2 \ll S_2/Q^2$,

$$|q_0(t_{\pm})^2 \delta q(t_{\pm})| \ll Q^2 \cdot \frac{S_2}{Q^2} = S_2.$$

Third,

$$|q_0(t_{\pm})(\delta q(t_{\pm}))^2| \ll Q \left(\frac{S_2}{Q^2} \right)^2 = \frac{S_2^2}{Q^3}.$$

Since from the tail-curvature theorem one already has $S_2 = o(Q)$, for Q sufficiently large this is $O(S_2/Q^2)$ and hence $\ll S_2$.

Finally,

$$|(\delta q(t_{\pm}))^3| \ll \left(\frac{S_2}{Q^2} \right)^3 = \frac{S_2^3}{Q^6} \ll S_2.$$

Therefore

$$|R_{22}| \ll S_2.$$

Collecting the four entrywise bounds, we obtain

$$|R_{11}|, |R_{12}|, |R_{21}|, |R_{22}| \ll S_2.$$

Since all norms on 2×2 matrices are equivalent,

$$\|R_{m,\pm}(s)\| \ll S_2,$$

uniformly on the microscopic disk. This proves the proposition. $\square$

Proposition 9.3 (Entry-by-entry perturbation estimate for the corrected mixed block). *Let*

$$q(t) = q_0(t) + \delta q(t), \quad \Phi(t) = \Phi_0(t) + \delta \Phi(t),$$

where q_0 is the baseline local-model-plus-affine-jet phase derivative and δq is the corrected remainder. Write

$$t_{\pm} = m \pm \frac{s}{2}, \quad \Delta(s) := \Phi(t_-) - \Phi(t_+), \quad \Delta_0(s) := \Phi_0(t_-) - \Phi_0(t_+),$$

and set

$$\delta \Delta(s) := \Delta(s) - \Delta_0(s).$$

Let $N_m^{\text{corr}}(s)$ be the corrected finite- s mixed block built from q, Δ , and let $N_m^{(0)}(s)$ be the corresponding block built from q_0, Δ_0 . Define

$$\tilde{R}_m(s) := N_m^{\text{corr}}(s) - N_m^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_{\pm})| \ll L_I^2 S_2, \quad |\delta q'(t_{\pm})| \ll L_I S_2, \quad |\delta q''(t_{\pm})| \ll S_2,$$

and

$$|\delta\Delta(s)| \ll L_I^3 S_2,$$

with

$$L_I \asymp \frac{x_0}{Q}, \quad |q_0(t_\pm)| \asymp Q, \quad |s| \asymp L_I.$$

Then

$$\|\tilde{R}_m(s)\| \ll S_2$$

uniformly for $|s| < \rho_0/Q$.

Proof. Write the corrected mixed block entrywise as

$$N_m^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta}{s} & \frac{\sin \Delta + q(t_+)s \cos \Delta}{s^2} \\ -\frac{\sin \Delta + q(t_-)s \cos \Delta}{s^2} & \frac{(q(t_-) + q(t_+))s \cos \Delta + (2 - q(t_-)q(t_+)s^2) \sin \Delta}{2s^3} \end{pmatrix},$$

and analogously for $N_m^{(0)}(s)$ with (q_0, Δ_0) .

We estimate the perturbation entry by entry. Throughout, use

$$\sin(\Delta_0 + \delta\Delta) - \sin \Delta_0 = O(|\delta\Delta|), \quad \cos(\Delta_0 + \delta\Delta) - \cos \Delta_0 = O(|\delta\Delta|).$$

(1,1)-entry.

$$\tilde{R}_{11} = -\frac{2}{\pi s} (\sin \Delta - \sin \Delta_0),$$

hence

$$|\tilde{R}_{11}| \ll \frac{|\delta\Delta|}{|s|} \ll \frac{L_I^3 S_2}{L_I} \ll L_I^2 S_2 \ll S_2.$$

(1,2)-entry. The numerator perturbation is

$$(\sin \Delta - \sin \Delta_0) + (q(t_+) - q_0(t_+))s \cos \Delta + q_0(t_+)s(\cos \Delta - \cos \Delta_0).$$

Therefore

$$|\tilde{R}_{12}| \ll \frac{|\delta\Delta| + |\delta q(t_+)| |s| + |q_0(t_+)| |s| |\delta\Delta|}{|s|^2}.$$

Using

$$|\delta\Delta| \ll L_I^3 S_2, \quad |\delta q(t_+)| \ll L_I^2 S_2, \quad |q_0(t_+)| \asymp Q, \quad |s| \asymp L_I \asymp Q^{-1},$$

we get

$$|\tilde{R}_{12}| \ll \frac{L_I^3 S_2}{L_I^2} + \frac{L_I^3 S_2}{L_I^2} + \frac{Q L_I^4 S_2}{L_I^2} \ll \frac{S_2}{Q} \ll S_2.$$

The same argument gives

$$|\tilde{R}_{21}| \ll S_2.$$

(2,2)-entry. Write the numerator as

$$F(q_-, q_+, \Delta; s) := (q_- + q_+)s \cos \Delta + (2 - q_- q_+ s^2) \sin \Delta.$$

Then

$$\tilde{R}_{22} = \frac{F(q(t_-), q(t_+), \Delta; s) - F(q_0(t_-), q_0(t_+), \Delta_0; s)}{2\pi s^3}.$$

A first-order expansion of F in (q_-, q_+, Δ) gives

$$|F - F_0| \ll |s| |\delta q(t_-)| + |s| |\delta q(t_+)| + |s|^2 Q |\delta q(t_-)| + |s|^2 Q |\delta q(t_+)| + (|s|Q + 1) |\delta\Delta|.$$

Using

$$|\delta q(t_\pm)| \ll L_I^2 S_2, \quad |\delta\Delta| \ll L_I^3 S_2, \quad |s| \asymp L_I \asymp Q^{-1},$$

we obtain

$$|F - F_0| \ll L_I^3 S_2.$$

Hence

$$|\tilde{R}_{22}| \ll \frac{L_I^3 S_2}{|s|^3} \ll S_2.$$

Collecting the four entrywise bounds and using equivalence of norms on 2×2 matrices gives

$$\|\tilde{R}_m(s)\| \ll S_2,$$

uniformly on the microscopic disk. $\square$

Lemma 9.4 (Baseline same-point variation). *Let $G_{m,\pm}^{(0)}(s)$ be the corrected same-point baseline blocks built from the explicit local model together with the affine jet $J_1[g_{\text{sm}}]$. Then, on the microscopic disk $|s| < \rho_0/Q$,*

$$\sup_{|s| < \rho_0/Q} \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \ll \rho_0 Q.$$

Proof. Differentiate the explicit same-point block formula entrywise:

$$G_{m,\pm}^{(0)}(s) = \frac{1}{\pi} \begin{pmatrix} 2q_0(t_{\pm}) & \frac{1}{2}q'_0(t_{\pm}) \\ \frac{1}{2}q'_0(t_{\pm}) & \frac{1}{12}(q''_0(t_{\pm}) + 2q_0(t_{\pm})^3) \end{pmatrix}, \quad t_{\pm} = m \pm \frac{s}{2}.$$

By the chain rule,

$$\partial_s G_{m,\pm}^{(0)}(s) = \pm \frac{1}{2\pi} \begin{pmatrix} 2q'_0(t_{\pm}) & \frac{1}{2}q''_0(t_{\pm}) \\ \frac{1}{2}q''_0(t_{\pm}) & \frac{1}{12}(q'''_0(t_{\pm}) + 6q_0(t_{\pm})^2 q'_0(t_{\pm})) \end{pmatrix}.$$

For the explicit local model plus affine jet, one has the baseline local scales

$$|q_0(t_{\pm})| \ll Q, \quad |q'_0(t_{\pm})| \ll Q^2, \quad |q''_0(t_{\pm})| \ll Q^3, \quad |q'''_0(t_{\pm})| \ll Q^4,$$

uniformly on the admissible midpoint class. The cubic combination $q'''_0(t_{\pm}) + 6q_0(t_{\pm})^2 q'_0(t_{\pm})$ therefore has size $O(Q^2)$ in the current normalization, and hence

$$\sup_{|s| < \rho_0/Q} \|\partial_s G_{m,\pm}^{(0)}(s)\| \ll Q^2.$$

Integrating along the segment from 0 to s gives

$$\|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \leq |s| \sup_{|u| < \rho_0/Q} \|\partial_u G_{m,\pm}^{(0)}(u)\| \ll \frac{\rho_0}{Q} \cdot Q^2 = \rho_0 Q.$$

$\square$

Proposition 9.5 (Corrected finite- s holomorphic whitening). *The corrected same-point and mixed blocks are holomorphic on*

$$|s| < \rho_0/Q.$$

Moreover, after shrinking ρ_0 if necessary, the same-point blocks

$$G_{m,\pm}^{\text{corr}}(s)$$

remain uniformly positive definite on this disk, and the inverse square roots

$$G_{m,\pm}^{\text{corr}}(s)^{-1/2}$$

are holomorphic there. Consequently

$$\hat{\Omega}_{\zeta}^{\text{corr}}(s; m) = G_{m,-}^{\text{corr}}(s)^{-1/2} N_m^{\text{corr}}(s) G_{m,+}^{\text{corr}}(s)^{-1/2}$$

is holomorphic on $|s| < \rho_0/Q$.

Proof. Holomorphy of the omitted smooth part and its required t -derivatives follows from Proposition 9.1. The only apparent poles at $s = 0$ in the corrected mixed block come from powers of $1/s$, $1/s^2$, $1/s^3$, and these are removable because $\Delta_m(s)$ is odd in s , $\sin \Delta_m(s) = O(s)$, and the off-diagonal and $(2, 2)$ numerators vanish to orders 2 and 3, respectively. Thus both $G_{m,\pm}^{\text{corr}}(s)$ and $N_m^{\text{corr}}(s)$ are holomorphic on the disk.

By Proposition 9.2 and the tail-curvature theorem,

$$\sup_{|s| < \rho_0/Q} \|R_{m,\pm}(s)\| = o(Q).$$

By Lemma 9.4,

$$\sup_{|s| < \rho_0/Q} \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \ll \rho_0 Q.$$

At $s = 0$, the baseline jet-limit block has spectral gap

$$\lambda_{\min}(G_{m,\pm}^{(0)}(0)) \gg Q$$

on the admissible midpoint class. Therefore

$$\|G_{m,\pm}^{\text{corr}}(s) - G_{m,\pm}^{(0)}(0)\| \leq \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| + \|R_{m,\pm}(s)\| \ll \rho_0 Q + o(Q).$$

For ρ_0 sufficiently small and m sufficiently large, this is smaller than half the baseline spectral gap, hence

$$\lambda_{\min}(G_{m,\pm}^{\text{corr}}(s)) \gg Q$$

uniformly on the disk. Holomorphic functional calculus then gives the holomorphic inverse square roots, and the whitened block is holomorphic as a product of holomorphic factors. $\square$

Proposition 9.6 (Corrected local deformation decomposition). *Let*

$$\Delta_\zeta(m, \sigma) := \widehat{\Omega}_\zeta^{\text{corr}}(s; m) - \widehat{\Omega}_{\text{bg}}^{\text{corr}}(s; m), \quad s = \sigma,$$

where $\widehat{\Omega}_{\text{bg}}^{\text{corr}}(s; m)$ denotes the corrected whitened block at background value-channel parameter $S(m) = 0$. Then

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

where $A_{\text{val}}(m, \sigma)$ is the pure value-channel derivative from (18), and

$$\|R_\zeta(m, \sigma)\| \ll S(m)^2 + S_2 + \text{Tail}(m, \sigma) + \text{Err}(m, \sigma).$$

In particular, in any regime where

$$S_2 + \text{Tail} + \text{Err} \ll L(m)S(m)^2,$$

one has

$$\|R_\zeta(m, \sigma)\| \ll L(m)S(m)^2.$$

Proof. Consider the analytic map

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

By Proposition 9.5, this map is holomorphic on the open set where $G_\pm$ are uniformly positive definite.

Write the corrected blocks as

$$G_{m,\pm}^{\text{corr}}(s) = G_{m,\pm}^{\text{bg}}(s) + S(m) \dot{G}_{m,\pm}(s) + E_{m,\pm}(s),$$

$$N_m^{\text{corr}}(s) = N_m^{\text{bg}}(s) + S(m) \dot{N}_m(s) + \tilde{E}_m(s),$$

where the $\dot{G}_{m,\pm}, \dot{N}_m$ terms are the pure value-channel first variations and the error terms satisfy

$$\|E_{m,\pm}(s)\| + \|\tilde{E}_m(s)\| \ll S_2 + \text{Tail} + \text{Err}.$$

By Propositions 9.2 and 9.3, this error control is uniform on the microscopic disk.

Applying first-order Taylor expansion of $\mathcal{W}$ at the background point gives

$$\widehat{\Omega}_\zeta^{\text{corr}}(s; m) = \widehat{\Omega}_{\text{bg}}^{\text{corr}}(s; m) + S(m) D\mathcal{W}_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+] + R_\zeta(m, \sigma),$$

where the remainder satisfies

$$\|R_\zeta(m, \sigma)\| \ll S(m)^2 + \|E_{m,-}\| + \|\tilde{E}_m\| + \|E_{m,+}\|.$$

By definition of the pure value-channel derivative, the first-order coefficient is exactly $A_{\text{val}}(m, \sigma)$. This yields

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

with the claimed bound on R_ζ . $\square$

Proposition 9.7 (Whitened mixed-block transfer with preserved Q^{-3} scale). *Let*

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

Write the corrected finite- s blocks as

$$G_{\pm}^{\text{corr}}(s) = G_{\pm}^{(0)}(s) + R_{\pm}(s), \quad N^{\text{corr}}(s) = N^{(0)}(s) + \tilde{R}(s),$$

and set

$$\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m) := \mathcal{W}(G_-^{\text{corr}}, N^{\text{corr}}, G_+^{\text{corr}}), \quad \widehat{\Omega}_{\zeta}^{(0)}(s; m) := \mathcal{W}(G_-^{(0)}, N^{(0)}, G_+^{(0)}).$$

Assume:

(1) *the same-point perturbations satisfy*

$$\|R_{\pm}(s)\| \ll S_2;$$

(2) *the corrected mixed-block perturbation satisfies*

$$\|\tilde{R}(s)\| \ll S_2;$$

(3) *the baseline same-point blocks are uniformly positive definite with spectral scales*

$$G_{\pm}^{(0)}(s) = \begin{pmatrix} O(Q) & O(1) \\ O(1) & O(Q^3) \end{pmatrix},$$

and hence

$$G_{\pm}^{(0)}(s)^{-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix};$$

(4) *the baseline mixed block has entry scales*

$$N^{(0)}(s) = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix};$$

(5) $S_2 = o(Q)$.

Then

$$\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m) - \widehat{\Omega}_{\zeta}^{(0)}(s; m) = \mathcal{T}_-(s; m) + \mathcal{T}_{\text{mix}}(s; m) + \mathcal{T}_+(s; m) + \mathcal{E}_2(s; m),$$

where

$$\mathcal{T}_-(s; m) := X_-(s) N^{(0)}(s) G_+^{(0)}(s)^{-1/2},$$

$$\mathcal{T}_{\text{mix}}(s; m) := G_-^{(0)}(s)^{-1/2} \tilde{R}(s) G_+^{(0)}(s)^{-1/2},$$

$$\mathcal{T}_+(s; m) := G_-^{(0)}(s)^{-1/2} N^{(0)}(s) X_+(s),$$

with

$$X_{\pm}(s) := D(G^{-1/2})_{G_{\pm}^{(0)}(s)}[R_{\pm}(s)],$$

and $\mathcal{E}_2$ quadratic in $(R_-, \tilde{R}, R_+)$.

Moreover,

$$\Phi_K(\mathcal{T}_-(s; m)) \ll \frac{S_2}{Q^3}, \quad \Phi_K(\mathcal{T}_{\text{mix}}(s; m)) \ll \frac{S_2}{Q^3}, \quad \Phi_K(\mathcal{T}_+(s; m)) \ll \frac{S_2}{Q^3},$$

and

$$\Phi_K(\mathcal{E}_2(s; m)) \ll \frac{S_2^2}{Q^3}.$$

Hence

$$\boxed{\Phi_K(\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m) - \widehat{\Omega}_{\zeta}^{(0)}(s; m)) \ll \frac{S_2}{Q^3}.}$$

Proof. The whitening map admits the first-order expansion

$$\widehat{\Omega}_\zeta^{\text{corr}} - \widehat{\Omega}_\zeta^{(0)} = D\mathcal{W}_{(0)}[R_-, \widetilde{R}, R_+] + \mathcal{E}_2,$$

where

$$D\mathcal{W}_{(0)}[R_-, \widetilde{R}, R_+] = X_- N^{(0)} G_+^{(0)-1/2} + G_-^{(0)-1/2} \widetilde{R} G_+^{(0)-1/2} + G_-^{(0)-1/2} N^{(0)} X_+.$$

We estimate the three first-order pieces separately.

Step 1: the mixed-block term. By the corrected mixed-block perturbation estimate,

$$\widetilde{R}_{11} \ll \frac{S_2}{Q^2}, \quad \widetilde{R}_{12}, \widetilde{R}_{21} \ll \frac{S_2}{Q}, \quad \widetilde{R}_{22} \ll S_2.$$

Using the baseline inverse-square-root scales

$$G_\pm^{(0)}(s)^{-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix},$$

the off-diagonal entries of

$$\mathcal{T}_{\text{mix}} := G_-^{(0)-1/2} \widetilde{R} G_+^{(0)-1/2}$$

are all $O(S_2/Q^3)$. Hence

$$\Phi_K(\mathcal{T}_{\text{mix}}) \ll \frac{S_2}{Q^3}.$$

Step 2: the same-point Fréchet derivative. We prove the corresponding estimate for the left same-point term

$$\mathcal{T}_- := X_- N^{(0)} G_+^{(0)-1/2}, \quad X_- := D(G^{-1/2})_{G_-^{(0)}}[R_-].$$

The proof for $\mathcal{T}_+$ is identical.

Write one baseline same-point block in the form

$$G_0 = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a \asymp Q, \quad b = O(1), \quad c \asymp Q^3,$$

and one perturbation as

$$R = \begin{pmatrix} r_{11} & r_{12} \\ r_{12} & r_{22} \end{pmatrix}, \quad r_{11} \ll \frac{S_2}{Q^2}, \quad r_{12} \ll \frac{S_2}{Q}, \quad r_{22} \ll S_2.$$

Diagonalize

$$G_0 = U \Lambda U^\top, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2),$$

with U orthogonal. Since

$$a \asymp Q, \quad c \asymp Q^3, \quad b = O(1),$$

we have

$$\lambda_1 \asymp Q, \quad \lambda_2 \asymp Q^3.$$

Moreover, if U is a rotation by angle θ , then

$$\tan(2\theta) = \frac{2b}{c-a} = O(Q^{-3}),$$

hence

$$\theta = O(Q^{-3}), \quad U = I + O(Q^{-3}).$$

Set

$$H := U^\top R U.$$

A direct computation gives

$$\begin{aligned} H_{11} &= r_{11} + 2\theta r_{12} + O(\theta^2 r_{22}), \\ H_{12} &= r_{12} + \theta(r_{22} - r_{11}) + O(\theta^2 r_{12}), \\ H_{22} &= r_{22} - 2\theta r_{12} + O(\theta^2 r_{11}). \end{aligned}$$

Using

$$\theta = O(Q^{-3}), \quad r_{11} \ll \frac{S_2}{Q^2}, \quad r_{12} \ll \frac{S_2}{Q}, \quad r_{22} \ll S_2,$$

we obtain

$$H_{11} \ll \frac{S_2}{Q^2}, \quad H_{12} \ll \frac{S_2}{Q}, \quad H_{22} \ll S_2.$$

Now let $f(x) = x^{-1/2}$. For a diagonal matrix Λ , the Fréchet derivative is given entrywise by the Löwner formula

$$Df_\Lambda(H)_{ij} = f^{[1]}(\lambda_i, \lambda_j)H_{ij},$$

where

$$f^{[1]}(u, v) = \begin{cases} \frac{u^{-1/2} - v^{-1/2}}{u - v}, & u \neq v, \\ -\frac{1}{2u^{3/2}}, & u = v. \end{cases}$$

Since

$$\lambda_1 \asymp Q, \quad \lambda_2 \asymp Q^3,$$

we have

$$f^{[1]}(\lambda_1, \lambda_1) \asymp Q^{-3/2}, \quad f^{[1]}(\lambda_2, \lambda_2) \asymp Q^{-9/2}, \quad f^{[1]}(\lambda_1, \lambda_2) \asymp Q^{-7/2}.$$

Therefore, if

$$Y := Df_\Lambda(H),$$

then

$$\begin{aligned} Y_{11} &\ll Q^{-3/2} \cdot \frac{S_2}{Q^2} = \frac{S_2}{Q^{7/2}}, \\ Y_{12} &\ll Q^{-7/2} \cdot \frac{S_2}{Q} = \frac{S_2}{Q^{9/2}}, \\ Y_{22} &\ll Q^{-9/2} \cdot S_2 = \frac{S_2}{Q^{9/2}}. \end{aligned}$$

Since

$$X = U Y U^\top$$

and $U = I + O(Q^{-3})$, these scales persist in the original basis:

$$X_{11} \ll \frac{S_2}{Q^{7/2}}, \quad X_{12}, X_{21} \ll \frac{S_2}{Q^{9/2}}, \quad X_{22} \ll \frac{S_2}{Q^{9/2}}.$$

Step 3: the left same-point term $\mathcal{T}_-$. The baseline mixed-block scales are

$$N^{(0)} = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix}, \quad G_+^{(0)-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix}.$$

We estimate only the off-diagonal entries, since

$$\Phi_K(Y) = Y_{12} + Y_{21}.$$

For $(\mathcal{T}_-)_{12}$, every summand is $O(S_2/Q^3)$. For example,

$$X_{11}N_{12}(G_+^{-1/2})_{22} \ll \frac{S_2}{Q^{7/2}} \cdot Q^2 \cdot Q^{-3/2} = \frac{S_2}{Q^3},$$

and

$$X_{12}N_{22}(G_+^{-1/2})_{22} \ll \frac{S_2}{Q^{9/2}} \cdot Q^3 \cdot Q^{-3/2} = \frac{S_2}{Q^3}.$$

All other terms are smaller. Thus

$$(\mathcal{T}_-)_{12} \ll \frac{S_2}{Q^3}.$$

The same argument gives

$$(\mathcal{T}_-)_{21} \ll \frac{S_2}{Q^3},$$

and hence

$$\Phi_K(\mathcal{T}_-) \ll \frac{S_2}{Q^3}.$$

By symmetry, the same estimate holds for the right same-point term:

$$\Phi_K(\mathcal{T}_+) \ll \frac{S_2}{Q^3}.$$

Step 4: the quadratic remainder. Every term in $\mathcal{E}_2$ contains at least two perturbation factors from $(R_-, \tilde{R}, R_+)$, and whitening contributes only polynomial Q -weights. Therefore

$$\Phi_K(\mathcal{E}_2) \ll \frac{S_2^2}{Q^3}.$$

Since $S_2 = o(Q)$, this is lower order than S_2/Q^3 .

Combining the estimates for $\mathcal{T}_-$, $\mathcal{T}_{\text{mix}}$, $\mathcal{T}_+$, and $\mathcal{E}_2$ gives

$$\Phi_K(\hat{\Omega}_\zeta^{\text{corr}}(s; m) - \hat{\Omega}_\zeta^{(0)}(s; m)) \ll \frac{S_2}{Q^3},$$

as claimed. $\square$

10. ODD HOLOMORPHIC TRANSVERSE CHANNEL

Define the corrected transverse scalar

$$(20) \quad H_m(s) := \Phi_K(\hat{\Omega}_\zeta^{\text{corr}}(s; m)).$$

By symmetry under $s \mapsto -s$, this is odd. By Corollary 8.2, the leading value channel is annihilated by Φ_K , so only the curvature, tail, and corrected finite- s remainder channels contribute.

Proposition 10.1 (Boundary estimate). *On the microscopic boundary*

$$|s| = \rho_0/Q,$$

one has

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3}.$$

Proof. By Proposition 9.6, the corrected local deformation has the form

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma).$$

Applying Φ_K and using Corollary 8.2,

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0,$$

we obtain

$$H_m(s) = \Phi_K(\hat{\Omega}_\zeta^{\text{corr}}(s; m)) = \Phi_K(R_\zeta(m, \sigma)).$$

By Proposition 9.7, passing the corrected perturbation through whitening and then through Φ_K preserves the microscopic transverse gain:

$$\Phi_K(\hat{\Omega}_\zeta^{\text{corr}}(s; m) - \hat{\Omega}_\zeta^{(0)}(s; m)) \ll \frac{S_2}{Q^3}.$$

By the tail-curvature theorem and the corrected transfer bounds,

$$S_2 \ll L(m)S(m)^2$$

in the corrected regime. Therefore

$$|H_m(s)| \ll \frac{L(m)S(m)^2}{Q^3}.$$

This is the claimed boundary estimate. $\square$

Proposition 10.2 (Odd microscopic expansion). *The function $H_m(s)$ is odd and holomorphic on $|s| < \rho_0/Q$, and therefore admits an expansion*

$$H_m(z/Q^2) = \sum_{r \geq 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}}, \quad |z| < \rho_0,$$

with

$$|c_{2r+1}(m)| \ll_{\rho_0} L(m)S(m)^2 \rho_0^{-(2r+1)}.$$

Proof. Oddness follows from symmetric placement and the jet-basis swap symmetry:

$$\widehat{\Omega}_\zeta^{\text{corr}}(-s; m) = J_0 \widehat{\Omega}_\zeta^{\text{corr}}(s; m) J_0,$$

with

$$\Phi_K(J_0 X J_0) = -\Phi_K(X).$$

Holomorphy follows from Proposition 9.5, and the coefficient bounds are immediate from Cauchy's estimate applied to Proposition 10.1. $\square$

11. N -POINT ODD-MOMENT CANCELLATION

For $N \leq \rho_0 Q$, set

$$s_j = \frac{j}{Q^2}, \quad j = 1, \dots, N,$$

and let $a_j^{(N)}$ be the explicit odd-moment-canceling coefficients satisfying

$$\sum_{j=1}^N a_j^{(N)} = 1, \quad \sum_{j=1}^N a_j^{(N)} j^{2r+1} = 0 \quad (0 \leq r \leq N-2).$$

Define the corrected N -point transverse scalar

$$(21) \quad \Xi_\zeta^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_m(s_j).$$

Proposition 11.1 (Corrected N -point transverse estimate for the full corrected scalar). *Let*

$$H_m(s) := \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m))$$

be the full corrected transverse scalar. For $N = \lfloor \alpha Q \rfloor$ with $\alpha > 0$ sufficiently small, define

$$s_j = \frac{j}{Q^2}, \quad \Xi_\zeta^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_m(s_j).$$

Then

$$|\Xi_\zeta^{(N)}(m)| \ll L(m) S(m)^2 e^{-\delta Q}$$

for some $\delta > 0$.

Proof. By Proposition 9.5, the full corrected scalar

$$H_m(s) = \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m))$$

is holomorphic on $|s| < \rho_0/Q$. By Proposition 10.1, it satisfies

$$|H_m(s)| \ll_{\rho_0} \frac{L(m) S(m)^2}{Q^3} \quad (|s| = \rho_0/Q).$$

By Proposition 10.2, it is odd and admits the expansion

$$H_m(z/Q^2) = \sum_{r \geq 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}},$$

with

$$|c_{2r+1}(m)| \ll L(m) S(m)^2 \rho_0^{-(2r+1)}.$$

Applying the standard odd-moment cancellation with the coefficients $a_j^{(N)}$ kills the first $N-1$ odd terms of this full corrected expansion. The surviving moment factor is exponentially tame for $N = \alpha Q$, which gives

$$|\Xi_\zeta^{(N)}(m)| \ll L(m) S(m)^2 e^{-\delta Q}.$$

$\square$

We isolate the toy microscopic expansion in Appendix A; the only fact used in the main argument is the following N -point preservation statement.

Proposition 11.2 (Toy N -point transverse preservation). *Fix a compact interval*

$$D = [d_-, d_+] \subset (0, \infty),$$

and let

$$\Xi_{\text{toy}}^{(N)}(u, d) := \sum_{j=1}^N a_j^{(N)} \Phi_K(\Delta_{\text{toy}}(u, d; s_j)), \quad s_j = \frac{j}{Q^2},$$

where the coefficients satisfy

$$\sum_{j=1}^N a_j^{(N)} = 1.$$

Then, for $d \in D$ and $|u| < u_0(D)$,

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

uniformly in $d \in D$. In particular,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2$$

uniformly on compact d -ranges.

Proof. By Lemma A.1, one has

$$\Delta_{\text{toy}}(u, d; s_j) = u^2 M(d) + u^4 E_j(u, d), \quad \sup_{1 \leq j \leq N} \|E_j(u, d)\| \ll 1$$

uniformly for $d \in D$. Applying Φ_K gives

$$\Phi_K(\Delta_{\text{toy}}(u, d; s_j)) = u^2 \Phi_K(M(d)) + u^4 \Phi_K(E_j(u, d)).$$

Summing over j and using

$$\sum_{j=1}^N a_j^{(N)} = 1$$

yields

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + u^4 \sum_{j=1}^N a_j^{(N)} \Phi_K(E_j(u, d)).$$

Since Φ_K is bounded and $\sup_j \|E_j(u, d)\| \ll 1$, the error term is $O(u^4)$, uniformly for $d \in D$. This proves the expansion.

Because $\Phi_K(M(d)) \neq 0$ for $d > 0$, and is bounded away from 0 on every compact interval $D \subset (0, \infty)$, the final estimate

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2$$

follows. □

On the toy side, Proposition 11.2 gives

$$(22) \quad \Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

so on compact d -ranges,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2.$$

Using (19),

$$(23) \quad |\Xi_{\text{toy}}^{(N)}(u, d)| \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

12. FINAL CONTRADICTION

The calibrated N -point comparison now closes the argument.

Lemma 12.1 (Crude polynomial bound for the local value scale). *Let*

$$S(m) = \sum_{\rho} f_{\beta_{\rho}, \gamma_{\rho}}(m),$$

where

$$f_{\beta, \gamma}(m) = \frac{1 - \beta}{(1 - \beta)^2 + (2m - \gamma)^2} + \frac{\beta}{\beta^2 + (2m - \gamma)^2}.$$

Then

$$S(m) \ll Q^2, \quad Q \asymp \log |m|.$$

Consequently

$$L(m) \ll Q$$

and

$$B_{\zeta}(m)L(m)S(m) \ll Q^4.$$

Proof. By the zero-free region, for zeros at relevant height one has

$$\min(\beta, 1 - \beta) \gg \frac{1}{Q}.$$

Hence

$$f_{\beta, \gamma}(m) \ll \frac{Q}{1 + (2m - \gamma)^2}.$$

Partition the zeros into unit shells

$$\mathcal{S}_k := \{\rho : k \leq |2m - \gamma_{\rho}| < k + 1\}, \quad k = 0, 1, 2, \dots$$

Standard zero counting gives

$$\#\mathcal{S}_k \ll Q$$

for each unit shell near height m . Therefore

$$S(m) \ll \sum_{k \geq 0} \#\mathcal{S}_k \frac{Q}{1 + k^2} \ll Q \sum_{k \geq 0} \frac{Q}{1 + k^2} \ll Q^2.$$

Since

$$L(m) = \log\left(3 + |m| + \frac{1}{S(m)}\right),$$

we trivially have $L(m) \ll Q$, and the last estimate follows from $B_{\zeta}(m) \asymp Q$. $\square$

Theorem 12.2 (Final calibrated N -point contradiction). *No fixed off-line toy anomaly can occur. Consequently the Riemann Hypothesis holds.*

Proof. By Proposition 8.6, in the fixed small- x_0 regime one has

$$u^2 \asymp \frac{x_0}{B_{\zeta}(m)} S(m).$$

On the toy side, Proposition 11.2 gives

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

hence on compact d -ranges,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2 \asymp \frac{x_0}{B_{\zeta}(m)} S(m).$$

On the zeta side, Proposition 11.1 gives

$$|\Xi_{\zeta}^{(N)}(m)| \ll L(m)S(m)^2 e^{-\delta Q}.$$

By Lemma 12.1, one has $S(m) \ll Q^2$, $L(m) \ll Q$, and $B_\zeta(m)L(m)S(m) \ll Q^4$. Therefore

$$\frac{L(m)S(m)^2 e^{-\delta Q}}{(x_0/B_\zeta(m))S(m)} = \frac{B_\zeta(m)}{x_0} L(m)S(m) e^{-\delta Q} \ll Q^4 e^{-\delta Q} \rightarrow 0.$$

Hence for Q sufficiently large,

$$|\Xi_\zeta^{(N)}(m)| < |\Xi_{\text{toy}}^{(N)}(u, d)|.$$

This is impossible if the corrected local zeta deformation realizes the local off-line toy anomaly with parameter $d \in D$, because the calibrated N -point construction was set up so that the toy model is the jet-limit local avatar of an off-line zero configuration. Thus no off-line zero can occur. Since every violation of the Riemann Hypothesis would produce such an off-line local anomaly, all nontrivial zeros must lie on the critical line. Hence RH holds. $\square$

APPENDIX A. TOY MICROSCOPIC EXPANSION

Lemma A.1 (Uniform microscopic expansion of the toy local deformation). *Fix a compact interval*

$$D = [d_-, d_+] \subset (0, \infty),$$

and let

$$\Delta_{\text{toy}}(u, d; s)$$

denote the toy local deformation matrix in the symmetric microscopic regime, with off-line parameter u , separation parameter d , and microscopic node s .

Then there exist constants $\rho_{\text{toy}} > 0$ and $u_0(D) > 0$ such that for all

$$d \in D, \quad |s| < \rho_{\text{toy}}, \quad |u| < u_0(D),$$

one has the expansion

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

where $M(d)$ is the toy anomaly matrix and

$$\sup_{\substack{d \in D \\ |s| < \rho_{\text{toy}} \\ |u| < u_0(D)}} \|E(u, d; s)\| \ll 1.$$

In particular, for the microscopic nodes

$$s_j = \frac{j}{Q^2}, \quad j = 1, \dots, N,$$

with $N \leq \rho_{\text{toy}} Q^2$, one has uniformly

$$\Delta_{\text{toy}}(u, d; s_j) = u^2 M(d) + u^4 E_j(u, d), \quad \sup_j \|E_j(u, d)\| \ll 1.$$

Proof. The toy local deformation is obtained from an explicit finite-dimensional off-line perturbation of the toy block. By construction it depends smoothly (in fact analytically) on the parameters (u, s, d) in a neighborhood of

$$(u, s, d) = (0, 0, d_0), \quad d_0 \in D.$$

At $u = 0$, the deformation vanishes, and by the even symmetry of the off-line split there is no linear term in u . Hence the first nontrivial term is quadratic:

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d, s) + O(u^4),$$

uniformly on compact d -ranges and for $|s|$ in a fixed microscopic disk.

By the jet-limit computation for the toy model, the leading quadratic coefficient is independent of the microscopic node s and equals the toy anomaly matrix $M(d)$. Thus

$$M(d, s) \equiv M(d),$$

and the expansion becomes

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

with E uniformly bounded on compact parameter ranges. Restricting to the microscopic nodes $s_j = j/Q^2$ gives the stated uniform bound for $E_j(u, d)$. $\square$